

Multicalibration for Unbiased Model-Based Prevalence Estimation

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Significance

Researchers increasingly use AI systems, including large language models, as measurement devices to estimate how common a phenomenon is in a population. A critical but overlooked problem arises when the target population differs from the one on which the device was validated: standard calibration methods produce biased prevalence estimates. Because prevalence estimates inform policy decisions, scientific conclusions, and resource allocation, such bias can have serious downstream consequences. We show that multicalibration (requiring a device to be accurate not just on average but conditional on the input features) is sufficient for unbiased prevalence estimation under covariate shift, and that no standard calibration method provides this guarantee. Our result, established theoretically and confirmed by simulation and empirical applications, implies that measurement in general and the rapidly growing body of research using AI for measurement specifically is vulnerable to systematic bias which could be mitigated by enforcing multicalibration.

Abstract

Estimating the prevalence of a category in a population using imperfect measurement devices (diagnostic tests, classifiers, or large language models) is fundamental to science, public health, and platform governance. Standard approaches correct for known device error rates but assume these rates remain stable across populations. We show this assumption fails under covariate shift and that multicalibration, which enforces calibration conditional on the input features rather than just on average, is sufficient for unbiased prevalence estimation under such shift. Standard calibration and quantification methods fail to provide this guarantee. Our work connects recent theoretical work on fairness to a longstanding measurement problem spanning nearly all academic disciplines. A simulation confirms that standard methods exhibit bias growing with shift magnitude, while a multicalibrated estimator maintains near-zero bias. Two empirical applications—estimating employment prevalence across U.S. states using the American Community Survey, and classifying political texts across four countries using an LLM—demonstrate that multicalibration substantially reduces bias in practice, while highlighting that calibration data should cover the key feature dimensions along which target populations may differ.

Introduction

Large language models (LLMs) are increasingly used as measurement devices for estimating the prevalence of a category in a population. Researchers now routinely deploy LLMs as zero-shot classifiers to estimate the frequency of phenomena that previously required expensive manual annotation: examples include,

coding democracy indicators across countries (Weidmann et al. 2025), classifying protest events in news corpora (Overös et al. 2024), estimating party policy positions from manifestos across dozens of countries and languages (Benoit et al. 2026), categorizing open-ended survey responses at near-human accuracy (Mellon et al. 2024; Gilardi et al. 2023), extracting diagnostic attributes from pathology reports (Sushil et al. 2024), identifying goals-of-care discussions in clinical notes (Sibley et al. 2025), annotating art forms in auction records (Tojima and Yoshida 2025), and converting qualitative text into quantitative variables across multiple languages (Karjus et al. 2025). LLMs are also being deployed for content moderation¹ and as LLM “judges” to assess the quality and safety of AI systems (Zheng et al. 2023; Yuan et al. 2024).

These applications are typically validated by reporting discriminative performance: accuracy, AUC, or agreement with human annotations. Researchers then use or advocate for using validated LLMs as measurement devices across new populations, time periods, or subgroups. But strong discriminative performance does not guarantee accurate prevalence estimates. When the composition of the target population differs from the validation population, a model that separates positives from negatives well can still produce prevalence estimates that are substantially biased, because discrimination is invariant to the calibration errors that drive prevalence bias. This problem is invisible to standard validation practice and, as we show, can produce large bias even with strong classification performance.

Neither the confidence elicitation literature nor the quantification² literature solves this problem. Confidence elicitation methods — verbalized confidence (Tian et al. 2023), token log-probabilities (Kadavath et al. 2022), consistency sampling (Wang et al. 2023) — and post-hoc calibration (Oliveira et al. 2025) can improve the quality of an LLM’s scores, but they target *global* calibration: calibration on the population where they are evaluated. Empirical evidence confirms that such calibration does not transfer: it degrades 2–3× under language shift even when accuracy is preserved (Yang et al. 2023), and varies widely across tasks and domains (Ren et al. 2025). Quantification methods such as “Classify & Count”, “Adjusted Count”, and the Saerens-Latinne-Decaestecker EM algorithm (see González et al. (2017) for a detailed survey) attempt to correct for classifier errors directly, but rely on the assumption that error properties remain static across populations. Importance-weighted methods handle covariate shift in principle but require re-estimating density ratios for each new target population. None of these approaches provides a measurement device that can be validated once and then reliably applied to new populations.

We show that multicalibration — calibration conditional on the input features, not just on average — fills this gap. Building on Kim et al. (2022)’s “Universal Adaptability” result, we show that a multicalibrated device requires no target-specific estimation: it is calibrated once on source data and produces unbiased prevalence estimates on any target population whose features lie within the calibration support. This property is critical for LLM-based measurement, where a single device is typically applied to many populations without the opportunity to re-calibrate for each one. Importantly, multicalibration can be applied to any LLM output — from discrete Yes/No classifications (the standard practice in applied research) to continuous confidence scores — because it operates on the scores’ relationship to outcomes conditional on features, not on the scores’ intrinsic quality. Our results apply more generally to any model-based measurement device, but the LLM setting is where the practical need is most acute and the calibration problem most overlooked.

We review existing approaches to the quantification problem, connect the quantification task to the broader theory of domain adaptation via multicalibration, and apply those insights to two empirical applications: a controlled demonstration using a standard ML classifier to estimate employment prevalence under age distribution shift (American Community Survey), followed by the main application of classifying political texts across four countries using an LLM as a zero-shot measurement device.

¹<https://openai.com/index/using-gpt-4-for-content-moderation/>

²The task of estimating prevalence from imperfect classifiers is known as *quantification* in the machine learning literature; we use this term interchangeably with *prevalence estimation*.

Results

Standard calibration methods fail under covariate shift

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$ producing probabilistic predictions. The goal is to estimate the population prevalence $\pi = P(Y = 1)$ in a target population using only unlabeled target data $\{X_i\}$ and the device h . The key assumption concerns which aspect of the data-generating process changes across populations. Three types of distributional change are commonly distinguished (Storkey 2009):

- **Covariate shift:** The marginal distribution of features $P(X)$ changes, but the conditional $P(Y | X)$ remains stable. This is natural when features causally influence outcomes ($X \rightarrow Y$): applying a device to a new population changes the mix of features but not the underlying feature-outcome relationships. Many measurement applications fit this pattern: demographics influence employment, text content determines topic relevance, patient characteristics drive diagnoses. However, some settings are better described by $Y \rightarrow X$ (for example, fraudulent intent may drive user behavior on a platform), in which case label shift is the more appropriate assumption.
- **Label shift** (prior probability shift): The prevalence $P(Y)$ changes, and the conditional $P(X | Y)$ remains stable. This is natural when outcomes generate features ($Y \rightarrow X$), as in some diagnostic settings where a disease produces symptoms. However, for most measurement tasks where the goal is to classify based on observed features, the $X \rightarrow Y$ direction is more appropriate.
- **Concept drift:** The conditional $P(Y | X)$ changes, not as a consequence of shifting $P(X)$ or $P(Y)$, but because the mechanism linking features to outcomes itself changes. For example, the meaning of words may evolve over time, or a disease may mutate to present differently. No purely statistical post-processing can correct for concept drift; new labeled data from the target distribution is required.

We focus on covariate shift, which is the dominant concern in the measurement settings motivating this paper (see Limitations for further discussion of concept drift).

A device is *globally calibrated* if $\mathbb{E}[Y | h(X) = p] = p$ for all prediction values p . Under global calibration, $\mathbb{E}[h(X)] = \mathbb{E}[Y] = \pi$, so the sample mean of predictions is an unbiased prevalence estimate. However, global calibration is a property of a *specific* population. A device calibrated on one population need not be calibrated on another, even under covariate shift with stable $P(Y | X)$.

The failure mechanism is straightforward. Suppose the population consists of subgroups G with weights w_G . Within each subgroup, the device may be biased: let $\epsilon_G = \mathbb{E}[Y - h(X) | X \in G]$ denote the mean prediction error within group G . Global calibration requires only that these errors cancel on average: $\sum_G w_G \epsilon_G = 0$. Under covariate shift, the group weights change to w_G^* , and the bias in the prevalence estimate becomes $\sum_G w_G^* \epsilon_G$, which is generally nonzero unless $\epsilon_G = 0$ for all G . Crucially, a device can be perfectly globally calibrated, with errors that precisely cancel in the training population, while having nonzero mean prediction error within every individual subgroup.

The quantification literature has proposed several methods to correct for device errors (González et al. 2017). Each is vulnerable to the failure mechanism described above. *Classify & Count* binarizes predictions at a threshold τ and reports the positive fraction; the threshold is chosen to match true prevalence on calibration data, but this correspondence breaks under shift. The *Adjusted Count* (Rogan-Gladen) estimator $\hat{\pi} = (\hat{p} - \text{FPR}) / (\text{TPR} - \text{FPR})$ uses sensitivity and specificity from calibration data (Rogan and Gladen 1978). These error rates are weighted averages of group-specific rates; when group proportions shift, the learned rates no longer match the target population's. The ratio estimator further amplifies errors when $\text{TPR} - \text{FPR}$ is small. *Probabilistic Adjusted Classify & Count* (PACC) generalizes

this to soft scores, using $\mathbb{E}[h(X) \mid Y=1]$ and $\mathbb{E}[h(X) \mid Y=0]$ in place of binary error rates, but fails for the same reason: these conditional means are weighted averages over subgroups and shift with population composition.

The *Saerens-Latinne-Decaestecker* (SLD) algorithm (Saerens et al. 2002), also known as EMQ, uses expectation-maximization to iteratively re-estimate prevalence by adjusting posteriors for a new prior; it is the strongest standard quantification method under label shift (where $P(X \mid Y)$ is stable), but under covariate shift, the class-conditional score distributions $P(h(X) \mid Y)$ change, causing SLD to converge to a biased estimate. The same argument applies to calibration approaches that assume stable error properties, such as the Calibrate-Extrapolate framework (Wu and Resnick 2024). Indeed, Wu and Resnick (2024) explicitly demonstrate that “covariate shift does not imply a stable calibration curve”: the calibrated posterior $P(Y=1 \mid h(X)=v)$ depends on the composition of individuals mapped to each score value v , which changes when $P(X)$ shifts. Their recommended estimator for the covariate-shift setting, averaging calibrated posteriors via isotonic regression, is precisely the global calibration baseline we evaluate below; as our results show, it still produces substantial bias because global calibration does not guarantee feature-conditional accuracy.

Multicalibration guarantees unbiased prevalence estimation

The analysis above shows that all standard methods produce biased prevalence estimates under covariate shift, with bias $\sum_G w_G^* \epsilon_G$ that is generally nonzero unless every $\epsilon_G = 0$. The illustration used discrete groups for clarity, but the insight generalizes to general conditioning on the features X .

Multicalibration (Hébert-Johnson et al. 2018) formalizes this requirement. A predictor $f(X)$ is *multicalibrated* with respect to a collection of subgroups $\mathcal{G}$ if $\mathbb{E}[Y \mid f(X) = v, X \in G] = v$ for every $G \in \mathcal{G}$ and prediction value v . When $\mathcal{G}$ is rich enough to capture all relevant structure in X , this is equivalent to requiring calibration conditional on the full feature space:

$$\mathbb{E}[Y \mid f(X) = v, X = x] = v$$

for all x and v with sufficient probability mass. This is the framing of Kim et al. (2022)’s “Universal Adaptability”: a predictor calibrated conditional on X yields correct expected values under any reweighting of $P(X)$, without requiring knowledge of the shift.

The connection to prevalence estimation is direct. If f is calibrated conditional on X , then $\mathbb{E}[f(X) \mid X = x] = \mathbb{E}[Y \mid X = x]$ for all x . Under covariate shift, where only $P(X)$ changes while $P(Y \mid X)$ remains stable, the law of iterated expectations gives (where $\mathbb{E}^*$ and P^* denote expectations and probabilities under the target distribution, and $\pi^* = P^*(Y=1)$ is the target prevalence):

$$\mathbb{E}^*[f(X)] = \int \mathbb{E}[f(X) \mid X = x] dP^*(x) = \int \mathbb{E}[Y \mid X = x] dP^*(x) = \mathbb{E}^*[Y] = \pi^*.$$

Because predictions are correct at each point in the feature space, the average of predictions tracks the true prevalence under *any* change in $P(X)$. Returning to the discrete illustration from the previous section: global calibration only ensures $\sum_G w_G \epsilon_G = 0$ for the training weights w_G ; calibration conditional on X ensures $\epsilon_G \approx 0$ everywhere, making the estimate robust to arbitrary changes in the feature distribution.

The condition that is strictly necessary for this robustness is *multi-accuracy*: $\mathbb{E}[f(X) - Y \mid X \in G] = 0$ for all groups G that the shift can reweight. This requires only that the *mean* prediction equals the mean outcome within each subgroup, not that the full calibration curve is correct within each subgroup. Multicalibration, which additionally requires $\mathbb{E}[Y \mid f(X) = v, X \in G] = v$, is a strictly stronger condition that implies multi-accuracy. In practice, multicalibration is the preferable target for two reasons. First,

practical algorithms like MCGrad naturally produce multicalibrated predictors, providing the stronger guarantee at no additional cost. Second, multicalibration protects against distribution shifts that are mediated by the scoring function itself. When the shift is a function of $f(X)$ (for example, when the device’s scores are used to guide importance sampling for label collection, or when a downstream decision rule selects units with $f(X) > \tau$), the reweighting operates along the score axis rather than along features X , and multi-accuracy with respect to X no longer suffices. Full calibration, $\mathbb{E}[Y \mid f(X) = v] = v$, is needed to correct for such score-dependent selection, and multicalibration provides this. Both conditions still require that the calibration features capture the dimensions along which the population shift occurs (an ignorability assumption); neither eliminates the need for thoughtful feature specification.

Practical post-hoc multicalibration algorithms can be applied to any base predictor, including LLMs and black-box classifiers, by iteratively correcting predictions in regions of the feature space where they are miscalibrated (Hébert-Johnson et al. 2018; Gopalan et al. 2022). Early algorithms required discrete groups to be specified explicitly, but more recent methods take a set of features as input and automatically discover miscalibrated regions, effectively approximating calibration conditional on the full feature space with increasing data (Tax et al. 2026)³.

Simulation: standard methods fail, multicalibration succeeds

We illustrate the theoretical results above with a simulation. We generate data with a binary covariate $X \in \{0, 1\}$ and binary outcome Y , where $P(Y = 1 \mid X = 1) = 0.85$ and $P(Y = 1 \mid X = 0) = 0.15$. A classifier produces systematically biased predictions: 10% underestimation when $X = 0$ and 10% overestimation when $X = 1$. All calibration parameters are learned on a balanced training distribution ($P(X = 0) = 0.5$). Prevalence is estimated by averaging the predicted probabilities over the target sample. We then evaluate prevalence estimates on test distributions where $P(X = 0)$ ranges from 0.01 to 0.99, repeating 50 times (details in Materials and Methods).

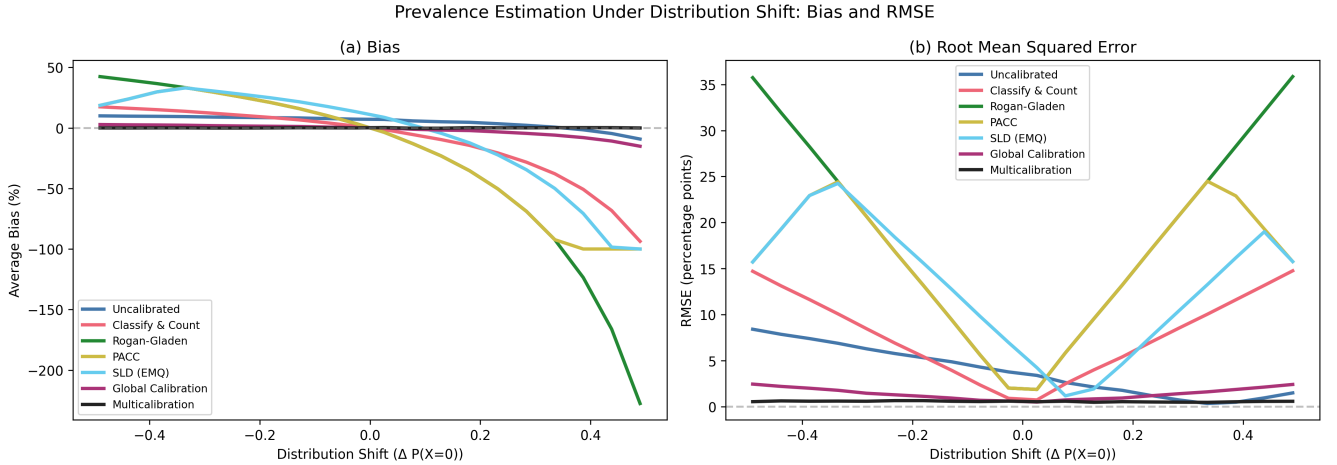


Figure 1: Figure 1

Figure 1: Prevalence estimation under covariate shift for six methods across 50 simulation runs. The x-axis shows $\Delta P(X=0)$, the change in $P(X=0)$ from the training value of 0.5. (a) Average bias, defined as the relative error $100 \times (\hat{\pi} - \pi) / \pi$ (%), averaged over 50 runs. (b) Root mean squared error (RMSE). Only multicalibration maintains near-zero bias and RMSE across all levels of distribution shift.

Figure 1 shows the results. At the training distribution (center), all methods except the uncalibrated baseline produce approximately unbiased estimates. As the distribution shifts, the methods diverge.

³An open-source implementation is available at <https://mcgrad.dev>.

Classify & Count and Rogan-Gladen show rapidly growing bias, exceeding $\pm 40\%$ at extreme shifts. The Rogan-Gladen estimator is particularly unstable: its ratio structure amplifies estimation errors, producing bias that exceeds the other methods by an order of magnitude. SLD tracks the Rogan-Gladen curve closely; its EM procedure converges to a similarly biased estimate because the class-conditional score distributions, which SLD assumes are stable, shift under covariate shift.⁴ Global calibration shows more moderate but still substantial bias (up to $\pm 15\%$). The RMSE panel (Figure 1b) shows that these bias differences dominate total estimation error: the RMSE curves closely track the absolute bias, confirming that variance is small relative to bias for all methods at this sample size. The multicalibrated estimator maintains near-zero bias and the lowest RMSE across the entire range of distribution shifts.

Empirical application: employment prevalence under age distribution shift

Before applying multicalibration to LLM-generated scores, which present additional challenges such as coarse discretization and non-standard score distributions, we first demonstrate the core mechanism in a controlled setting with a standard ML classifier. We analyze data from the American Community Survey (ACS), a large-scale annual survey conducted by the U.S. Census Bureau. The concept to be measured is the rate of employment. The measurement device is a logistic regression model of binary employment status, with 16 sociodemographic features including age, education, marital status, disability status, and citizenship.

Setup. We train a logistic regression classifier on data from eight U.S. states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018, totaling approximately 1.5 million training observations. The remaining in-distribution data is split into a calibration set ($n \approx 644,000$) and a test set ($n \approx 920,000$). Six additional states (CA, NY, FL, WA, AZ, CO) are held out for out-of-distribution (OOD) evaluation. We fit two post-hoc calibration methods on the calibration set: isotonic regression (global calibration) and MCGrad, a multicalibration algorithm that enforces calibration conditional on both categorical and numerical features. We compare seven prevalence estimation methods: raw (uncalibrated) scores, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, PACC, SLD, isotonic regression, and MCGrad.

Age distribution shift. Employment rates vary dramatically by age: approximately 47% for ages 16–24, 76% for ages 25–54, 61% for ages 55–64, and 17% for ages 65+. This makes age an ideal dimension along which to construct meaningful distribution shifts. We create synthetic target populations by resampling test data with shifted age distributions: young-skewed (oversampling ages 16–30), old-skewed (oversampling ages 60+), and bimodal (oversampling both tails). The resulting populations have true employment rates ranging from 12.8% to 46.0%. All calibration parameters are estimated once on the original calibration set and held fixed across scenarios.

Figure 2: Prevalence estimation bias (percentage points) under synthetic age distribution shift, for in-distribution data (left) and out-of-distribution states (right).

Setting	Age Dist.	True	Raw	CC	RG	PACC	SLD	IPW	Iso.	MCGrad
		Prev.								
In-Dist	Original	46.0%	-0.31	-0.07	+0.26	-0.07	+0.01	-0.3	-0.30	-0.27
In-Dist	Young-skewed	12.8%	+1.93	+2.47	-12.82	-12.82	-11.95	-0.3	+2.04	-0.11

⁴SLD’s bias is non-monotonic in the magnitude of shift: it grows at moderate shifts but decreases slightly at the extremes. This occurs because SLD’s output is bounded in $[0, 1]$ (as a mean of adjusted posteriors), so it saturates at 0 or 1 under extreme shift, while the true prevalence asymptotes to 0.15 or 0.85. The gap between these limits is smaller than the gap at intermediate shifts where SLD has not yet saturated.

Setting	Age Dist.	True Prev.	Raw	CC	RG	PACC	SLD	IPW	Iso.	MCGrad
In-Dist	Old-skewed	16.8%	+7.23	-6.62	-16.77	-16.77	-16.76	-1.2	+6.65	+0.22
In-Dist	Bimodal	21.1%	+4.57	-1.50	-18.62	-19.97	-16.14	+4.7	+4.33	+0.12
OOD	Original	45.1%	+1.15	+1.40	+2.12	+2.13	+2.25	+1.7	+1.17	+1.35
OOD	Young-skewed	13.0%	+2.93	+3.73	-12.96	-12.96	-11.38	+0.8	+3.08	+0.88
OOD	Old-skewed	16.0%	+8.47	-5.64	-15.97	-15.97	-15.97	+0.1	+7.91	+1.01
OOD	Bimodal	20.8%	+5.91	+0.09	-16.14	-17.27	-15.20	+6.3	+5.69	+1.13

Table 1: Prevalence estimation bias in percentage points (pp) under synthetic age distribution shift. IPW = importance-weighted prevalence estimation (target-specific density ratio via logistic regression). RG = Rogan-Gladen, Iso. = Isotonic regression.

Results. With no age shift, all methods produce approximately unbiased estimates (Table 1). Under shift, the methods diverge sharply.

Rogan-Gladen exhibits catastrophic failure, with bias of -12.8 to -18.6 percentage points in the in-distribution setting, effectively estimating zero or negative prevalence when the true rate is 13–21%. PACC and SLD show comparable bias (-12 to -20pp and -11 to -16pp respectively), confirming the theoretical prediction: all three methods rely on calibration quantities (binary error rates, conditional score means, or class-conditional score distributions) that shift with population composition. Isotonic regression and raw scores show more moderate but still substantial bias of 2–8 percentage points, reflecting the instability of global calibration curves under compositional change.

IPW, the standard covariate-shift baseline, performs well on simple shifts (young-skewed: -0.3pp, old-skewed: -1.2pp) but shows substantial bias on the bimodal shift (+4.7pp in-distribution, +6.3pp OOD). The bimodal shift creates a complex, non-monotonic density ratio that logistic regression does not capture well. IPW also requires re-estimating density ratios for each target population, unlike MCGrad which is fitted once.

MCGrad produces near-zero bias across all in-distribution age-shift scenarios (≤ 0.27 pp), including the bimodal shift where IPW struggles. Because MCGrad is multicalibrated with respect to age and other features, its predictions are correct *conditional on* these covariates. When the age distribution shifts, the feature-conditional predictions remain valid, and their average tracks the true prevalence. Bootstrap resampling (200 iterations per scenario) confirms that MCGrad also achieves the lowest RMSE, indicating that the bias reduction does not come at the cost of increased variance.

In the OOD setting, where the model is applied to states never seen during training while also shifting the age distribution, MCGrad’s advantage persists with modestly larger bias (0.88–1.35pp), reflecting the additional geographic shift along a dimension not included in the calibration features. IPW shows comparable OOD performance on simple shifts but again fails on the bimodal shift (+6.3pp).

LLM-based topic classification under cross-national shift

The ACS application uses a standard ML classifier with well-distributed probability estimates. We now examine the setting that motivates this paper: using an LLM as a zero-shot measurement device across multiple countries and languages. We compare two LLM output modes that reflect current practice and the confidence elicitation literature, respectively: (1) discrete Yes/No classifications, which is how LLMs

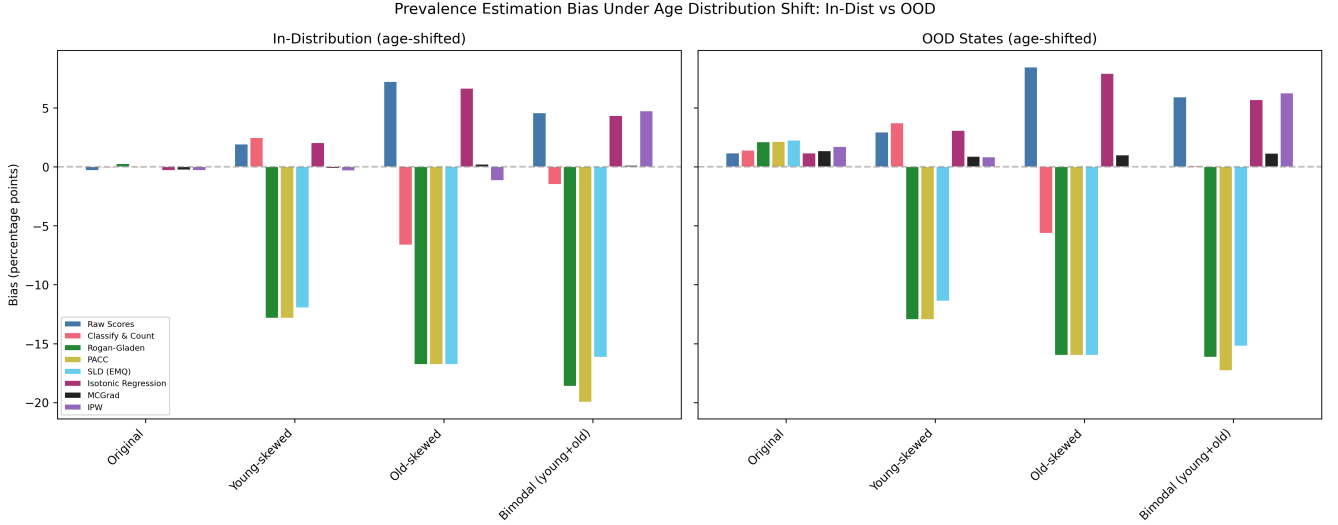


Figure 2: Figure 2

are used in all of the applied studies cited in the introduction, and (2) probability scores obtained via direct probability elicitation, where the LLM estimates $P(\text{Yes})$ and $P(\text{No})$ without first committing to an answer. This design lets us test whether confidence elicitation improves prevalence estimation, and whether multicalibration is needed in either case.

Setup. We use Claude Opus 4.6 to classify 30,000 political texts from the Comparative Agendas Project (Baumgartner et al. 2006) as related to Law & Crime (CAP major topic code 12). The data comprise six sub-populations across four countries and four languages (5,000 documents each): Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles, and Belgian TV news. Each document is classified twice in independent campaigns: once for a binary Yes/No label, and once for direct probability estimates $P(\text{Yes})$ and $P(\text{No})$ (summing to 1.0). The LLM achieves strong discriminative performance: AUC 0.959 for binary labels and 0.988 for probability scores on the in-distribution test set, with per-language AUCs of 0.978–0.985 for the probability scores.

The calibration set ($n \approx 13,400$) draws equally from four sub-populations (Denmark questions, Spain questions, U.S. bills, Belgium newspaper), ensuring that all four countries and languages are represented. Two sub-populations are held out as out-of-distribution targets: Spanish media and Belgian TV news, which share country and language with calibration data but introduce an unseen document type. MCGrad is calibrated with categorical features (country, document type, party) and two numerical features (decade, document length).

For the binary-label condition, MCGrad receives the LLM’s Yes/No classification as a categorical input feature, with all initial scores set to the calibration-set base rate. MCGrad then learns feature-conditional prevalence estimates from the label and metadata alone, without any probability score from the LLM. For the probability-score condition, MCGrad receives the LLM’s $P(\text{Yes})$ directly as the input score and calibrates it conditional on the same metadata features.

Results. Table 2 shows prevalence estimation bias across five scenarios: a baseline with no shift, two within-calibration shifts (country composition, document type composition), and two OOD scenarios.

Scenario	Shift Type	True Prev.	CC	RG	IPW	Iso.	MC (binary)	MC (scores)
Baseline	None	8.1%	+1.9	+0.5	-0.0	-0.3	-0.2	-0.2

Scenario	Shift Type	True Prev.	CC	RG	IPW	Iso.	MC (binary)	MC (scores)
Country shift	Within-cal.	8.9%	+2.5	+1.7	-0.4	+0.3	-0.5	-0.2
Doc-type shift	Within-cal.	6.8%	+1.6	-0.3	-0.4	-0.2	-0.2	+0.0
Spain media	OOD doc type	19.5%	+3.6	+3.1	-12.1	-2.7	-2.5	-4.4
Belgium TV	OOD doc type	11.1%	+4.8	+3.7	-4.4	+2.4	+0.4	+1.6

Table 2: Prevalence estimation bias (pp) for Law & Crime topic under cross-national and document-type shift. CC = Classify & Count (fraction of Yes labels); RG = Rogan-Gladen adjustment on binary labels; IPW = importance-weighted estimation (target-specific density ratio); Iso. = isotonic regression on probability scores; MC (binary) = MCGrad on binary labels with base-rate initialization; MC (scores) = MCGrad on probability scores. All methods except IPW are calibrated once on source data.

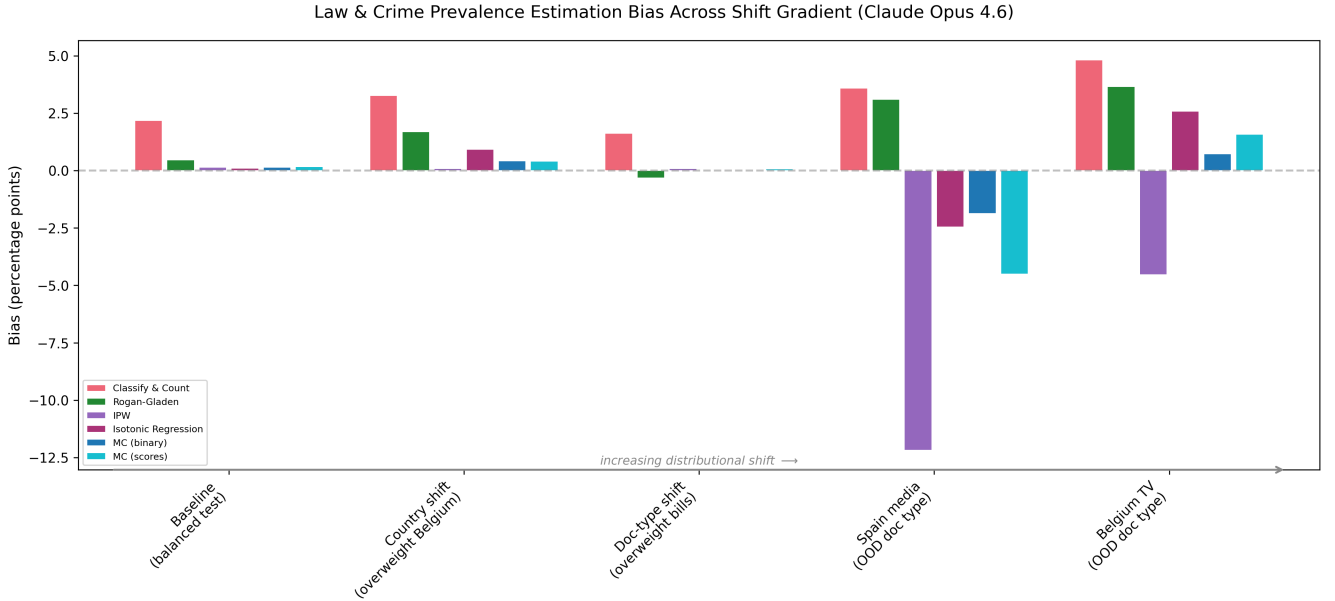


Figure 3: Figure 3

Figure 3: Prevalence estimation bias (pp) across the shift gradient. MCGrad achieves near-zero bias within the calibration distribution in both binary-label and probability-score conditions. On OOD populations (unseen document type), bias increases but remains smaller than all other methods.

The broken workflow is clearly visible: Classify & Count, the standard practice of counting positive LLM classifications, produces bias of +1.6 to +4.8pp across all scenarios. The Rogan-Gladen adjustment, the natural correction a practitioner might apply to binary labels, reduces bias within calibration but still shows +3.1 to +3.7pp on OOD populations.

MCGrad on binary labels achieves near-zero bias within the calibration distribution (≤ 0.5 pp) and degrades gracefully on OOD targets: +0.4pp on Belgian TV and -2.5pp on Spanish media. This is achieved using only the LLM’s Yes/No classification and document metadata, with no probability scores. MCGrad on probability scores achieves comparable within-calibration performance (≤ 0.2 pp) but shows larger

OOD bias on Spanish media (-4.4pp) and Belgian TV (+1.6pp). The finding that binary labels with MCGrad can match or outperform probability scores with MCGrad suggests that the metadata features, not the input score quality, drive the calibration improvement.

IPW, the standard covariate-shift method, achieves near-zero within-calibration bias but fails severely on OOD populations (-12.1pp on Spanish media), confirming the positivity violation observed in the ACS application. Unlike MCGrad, IPW requires re-estimating density ratios for each target population.

Comparison with the ACS application. Across both applications, the pattern is consistent: MCGrad achieves near-zero bias when the target population’s features are within the calibration support, and degrades when the shift is along an uncalibrated dimension. IPW matches MCGrad within calibration but fails harder on OOD targets. MCGrad’s practical advantage is that it requires no target-specific estimation: a single calibrated device can be applied to any target population. This is the “universal adaptability” property of Kim et al. (2022). We replicate the CAP analysis using Llama 3.3 70B Instruct (an open-weight model) in SI Appendix S2; the results are consistent, confirming that the findings are not specific to a particular LLM.

Discussion

We have shown that multicalibration, calibration conditional on the input features rather than just on average, is sufficient for accurate model-based prevalence estimation under population shift. The minimal theoretical requirement is multi-accuracy (correct predictions on average within each subgroup), which multicalibration implies. Both conditions require that the calibration features capture the relevant dimensions of population shift, and both require that the target population’s features lie within the support of the calibration distribution. Both empirical applications confirm this: when calibrated features overlap with the target population, MCGrad achieves near-zero bias; when the target introduces a novel feature value, bias increases with the severity of the shift. Practitioners should ensure their calibration data covers the key dimensions along which target populations may differ.

A central practical advantage of multicalibration over traditional covariate-shift methods is that it produces a *target-independent* measurement device. Importance-weighted methods achieve comparable within-calibration performance but require estimating density ratios for each new target population, and our CAP results show they fail more severely on OOD targets (IPW: -12.1pp on Spanish media, vs. MCGrad: -2.5pp) due to positivity violations when target features are outside the source support. A multicalibrated device, by contrast, is calibrated once and deployed without target-specific re-estimation. This is the “universal adaptability” property of Kim et al. (2022), and it is what makes multicalibration particularly well-suited to the measurement setting, where a single device is often applied to many target populations.

A notable finding from the CAP application is that MCGrad on discrete binary labels achieves comparable or better prevalence estimation than MCGrad on continuous probability scores. This suggests that for prevalence estimation under covariate shift, the metadata features that capture the shift structure matter more than the quality of the LLM’s confidence scores. This has immediate practical implications: researchers using the standard workflow of prompting an LLM for Yes/No classifications can apply multicalibration directly to those labels, without needing to extract probability scores via confidence elicitation methods.

The practical implications are immediate. The growing use of LLMs as zero-shot measurement devices rests on the assumption that models validated on one population can be reliably applied to others. Our analysis shows this assumption is safe only when the model is multi-accurate—and ideally multicalibrated—with respect to features that may shift across populations. Without this, prevalence estimates

in new populations, the primary use case for LLM-based measurement, are unreliable. Researchers using AI-based measurement should treat multicalibration not as an optional fairness criterion, but as a practical tool for ensuring valid inference.

This result also clarifies why certain widely-used practices are inadequate. The common approach of reporting a model’s overall accuracy, F1, or AUC on a held-out set provides no information about calibration, let alone multicalibration. A model with 95% accuracy can still produce heavily biased prevalence estimates if its errors are unevenly distributed across the feature space. Similarly, reporting a single calibration curve, as is standard in the quantification literature (González et al. 2017)—certifies only global calibration, which we have shown is insufficient under shift. Even comprehensive methodology textbooks for text-as-data research devote extensive attention to classification accuracy and validation but do not discuss calibration as a requirement for valid prevalence estimates (Grimmer et al. 2022); multicalibration is entirely absent from these discussions.

Applying multicalibration to LLM-based measurement. Multicalibration does not require access to a model’s internals. When LLMs classify documents, survey responses, or clinical notes, observable metadata such as document source, author demographics, text length, and language can serve directly as segment features, and these often correspond to the dimensions along which population composition may shift. For richer feature structure, Detommaso et al. (2024) demonstrate two approaches for constructing features over prompt-completion pairs: clustering within the embedding space and “self-annotation” via yes/no questions about input properties. Both yield features enabling effective multicalibration of LLM confidence scores, though end-to-end pipelines for LLM-based prevalence estimation remain an active area of development.

Several limitations warrant discussion. First, multicalibration requires specifying the features over which to calibrate; the guarantee holds only for shifts along included dimensions. As the CAP application demonstrates, performance degrades when target populations introduce novel feature values, particularly when compounded by a large prevalence difference. Practitioners should ensure calibration data covers the range of feature values likely to appear in target populations. Second, multicalibration requires labeled calibration data of sufficient size. This is feasible for standard ML pipelines but more challenging for zero-shot LLM settings, where calibration labels may require the manual annotation LLMs were intended to avoid; prediction-powered inference (Angelopoulos et al. 2023) offers a complementary framework. Third, LLMs produce text rather than probability scores; calibratable scores can be obtained via token log-probabilities, verbalized confidence elicitation (Tian et al. 2023) (as used in our CAP application), or repeated sampling (Detommaso et al. 2024). Fourth, our framework assumes covariate shift ($P(Y | X)$ stable). In settings where the causal direction is $Y \rightarrow X$ (such as fraud detection, where intent drives observable behavior), label shift may be a more appropriate assumption, and methods like SLD that are designed for label shift may be preferable. Under concept drift, no purely statistical correction substitutes for new labeled data. Finally, both applications use settings where ground-truth labels enable direct bias measurement—synthetic shifts in the ACS case and expert-coded labels across countries in the CAP case.

These results connect two literatures that have developed in isolation. The quantification literature has focused on correcting aggregate error rates but has not engaged with feature-conditional calibration (González et al. 2017; Wu and Resnick 2024). The multicalibration literature has focused on individual-level prediction quality and fairness but has not emphasized the implications for population-level inference (Hébert-Johnson et al. 2018; Kim et al. 2022). Our contribution is to show that calibration conditional on the feature space is the missing link that makes model-based prevalence estimation reliable under the distribution shifts that motivate its use.

Materials and Methods

Simulation. We generate $n = 10,000$ observations with a binary covariate $X \in \{0, 1\}$ and outcome $Y \sim \text{Bernoulli}(P(Y | X))$, where $P(Y = 1 | X = 1) = 0.85$ and $P(Y = 1 | X = 0) = 0.15$. The classifier produces deterministic scores $\hat{p}(X = 0) = 0.135$ and $\hat{p}(X = 1) = 0.935$, representing 10% multiplicative bias within each stratum. For each of $B = 50$ iterations, we generate fresh calibration data from $P(X = 0) = 0.5$, estimate all method-specific parameters, then evaluate bias and mean squared error on 20 test distributions with $P(X = 0)$ ranging from 0.01 to 0.99. The seven methods compared (un-calibrated averaging, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, PACC, SLD/EMQ, global multiplicative calibration, and multicalibration with stratum-specific additive corrections) are detailed with full mathematical definitions in the SI Appendix.

Empirical application (ACS). We use the ACS Public Use Microdata Sample via the *folktables* package, with the ACSEmployment prediction task (binary: employed vs. not employed) and 16 sociodemographic features. Training data comprises eight states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018. The base model is logistic regression with standard scaling. Post-hoc calibration uses isotonic regression (global) and MCGrad (multicalibration with categorical and numerical segment features) on a held-out calibration set ($n \approx 644,000$). In addition to raw scores, Classify & Count, Rogan-Gladen, isotonic regression, and MCGrad, we include two methods from the quantification literature: PACC (Probabilistic Adjusted Classify & Count), which generalizes Rogan-Gladen to continuous scores (González et al. 2017), and SLD (Saerens-Latinne-Decaestecker), an EM algorithm that iteratively adjusts posteriors for a new prior (Saerens et al. 2002). Synthetic age-shifted populations are created by importance-weighted resampling of the test set ($n \approx 920,000$), with exponential weights favoring young ages, old ages, or both extremes. RMSE is computed via bootstrap resampling (200 iterations per scenario). Six additional states (CA, NY, FL, WA, AZ, CO) serve as OOD evaluation data.

Empirical application (CAP). We use data from the Comparative Agendas Project (Baumgartner et al. 2006), which provides expert-coded policy topic labels for political texts across countries. The binary outcome is whether a document addresses Law & Crime (CAP major topic code 12). We sample 30,000 documents (5,000 from each of six sub-populations: Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles, and Belgian TV news). The measurement device is Claude Opus 4.6. Each document is classified in two independent campaigns: (1) binary Yes/No classification using the CAP codebook definition of Law & Crime, and (2) direct probability elicitation, where the LLM estimates $P(\text{Yes})$ and $P(\text{No})$ without first committing to an answer. The two campaigns were run independently to avoid anchoring contamination. The calibration set ($n \approx 13,400$) draws equally from four sub-populations (Denmark questions, Spain questions, U.S. bills, and Belgium newspaper). MCGrad is calibrated with categorical features (country, document type, political party) and two numerical features (decade and document length). For the binary-label condition, the LLM’s Yes/No classification is passed to MCGrad as a categorical input feature, with initial scores set to the calibration-set base rate; MCGrad learns feature-conditional prevalence estimates from the metadata and LLM label alone. For the probability-score condition, MCGrad receives $P(\text{Yes})$ as the input score. Isotonic regression is fitted on probability scores using the same calibration set. Rogan-Gladen adjustment uses binary-label TPR and FPR estimated on the calibration set. IPW estimates density ratios via logistic regression on country, document type, decade, and document length. Two sub-populations are held out as out-of-distribution targets: Spanish media and Belgian TV news, which differ from the calibration data only in document type while sharing the same countries and languages.

Software. MCGrad is available at github.com/facebookincubator/MCGrad. Simulation and analysis code are available at [TODO: insert repository URL before submission].

Author Contributions

[TODO: Add author contributions, e.g., “A.B. designed research; C.D. performed research; E.F. contributed analytic tools; A.B. and C.D. analyzed data; and A.B. wrote the paper.”]

Competing Interests

The authors declare no competing interests.

Data Availability

The American Community Survey data are publicly available via the *folktables* package. The Comparative Agendas Project data are publicly available at <https://www.comparativeagendas.net>. Simulation and analysis code are available at [TODO: insert repository URL].

Keywords

multicalibration, prevalence estimation, covariate shift, large language models, quantification

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